Oxford Cambridge and RSA – style practice paper

GCSE (9–1) Computer Science

J277/01 Computer systems

Topic 1.3: Networks, connections and protocols

Candidate forename		**Candidate surname**	
Centre number		**Candidate number**	

Practice questions in the style of OCR GCSE Computer Science (J277). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 100 minutes (roughly 1 minute per mark)
Instructions	• Use black ink. HB pencil may be used for graphs and diagrams only. • Answer **all** the questions. • Write your answer to each question in the space provided. • Where the question asks you to write code or pseudocode, you may use any suitable high-level language or the OCR Reference Language; marks are for the logic, not exact syntax. • Do **not** use a calculator or any other electronic device unless told to.
Information	• The total mark for this paper is **90**. • The marks for each question are shown in brackets [ ]. • Quality of extended responses will be assessed in questions marked with an asterisk (*). • Where a question asks you to *state* or *give*, a short answer is enough; *describe* and *explain* need more detail.

Name: _____ Date: _____ Mark: _____ / 90

1 A school has a computer network.

(a) State the difference between a **LAN** and a **WAN**. **[2]**

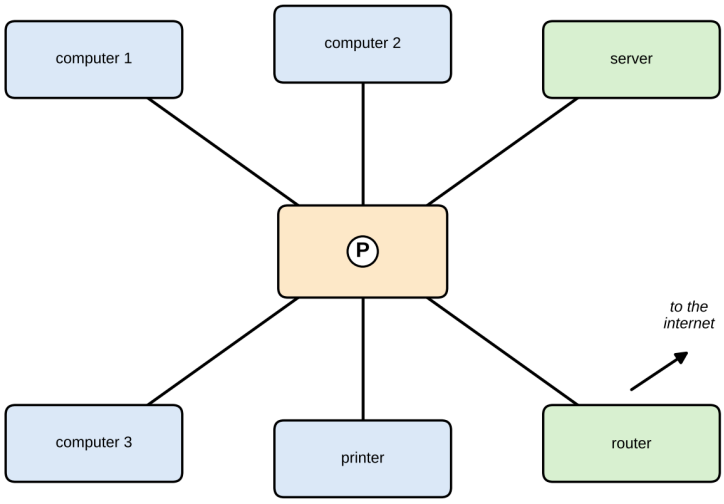
(b) Give **one** example of a WAN. **[1]**

(c) Give **two** benefits of connecting the school's computers to a network. **[2]**

(d) The network performance falls when many students log on at once. Give **two** factors that affect the performance of a network. **[2]**

(e) Explain how adding more users to a network can reduce its performance. **[2]**

2 The diagram shows a school computer room that uses a **star** network. Component **P** is at the centre.



(a) Name component **P**. **[1]**

(b) Give **two** advantages of a star topology compared to a bus topology. **[2]**

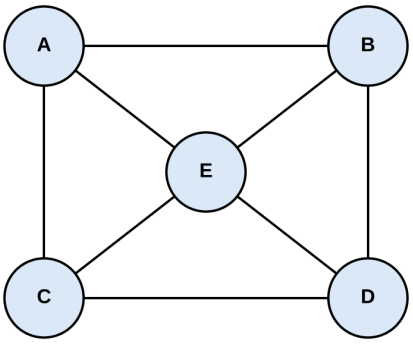
(c) Give **one** disadvantage of a star topology. **[1]**

(d) The router is connected to the switch. Describe the purpose of a **router**. **[2]**

(e) Name the hardware that each computer must contain to connect to the network. **[1]**

(f) The school plans to let students connect their own tablets wirelessly. Name the piece of hardware needed to allow this. **[1]**

3 The diagram shows five computers, **A** to **E**, connected in a **mesh** network.



(a) State how many direct links computer **E** has to other computers. **[1]**

(b) Give **two** advantages of a mesh topology. **[2]**

(c) Give **one** disadvantage of a mesh topology. **[1]**

(d) Computer **A** sends data to computer **D**. The direct link between **A** and **E** is damaged. Describe one route the data could take. **[2]**

4 Networks can use wired or wireless connections.

(a) State **two** advantages of a wireless network compared to a wired network. **[2]**

(b) State **two** disadvantages of wireless connections compared to wired. **[2]**

(c) Name **one** wireless technology used to connect a phone to wireless headphones. **[1]**

(d) Name **one** type of cable used in wired networks. **[1]**

(e) A company has an office building where data must be sent very quickly between servers. Recommend a transmission medium and justify your answer. **[3]**

(f) Explain how encryption helps to keep data sent over a wireless network secure. **[3]**

5 Computers on a network use addresses to identify them.

(a) Tick (✓) **one** box to show which of these is an example of an IPv4 address. **[1]**

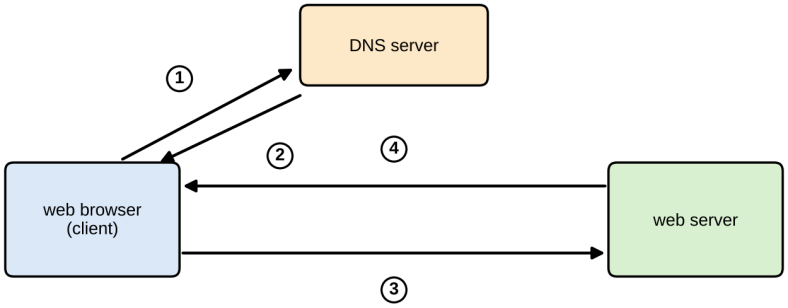
☐	00:1B:44:11:3A:B7
☐	192.168.1.24
☐	2001:0db8:85a3::8a2e
☐	www.example.com

(b) Give **one** difference between an IP address and a MAC address. **[1]**

(c) Explain why IPv6 was introduced. **[2]**

(d) State the purpose of a **MAC address**. **[1]**

(e) The domain name **www.example.com** is typed into a browser. The diagram shows the process that follows. Describe what happens at each numbered step. **[4]**



6 Protocols are sets of rules that devices follow when communicating.

(a) State what is meant by a **protocol**. **[1]**

(b) Draw **one** line from each protocol to its correct use. **[5]**

Protocol	**Use**
HTTP	Sending email to a mail server
HTTPS	Accessing email stored on a server
FTP	Transferring web pages in plain text
SMTP	Transferring web pages securely
IMAP	Transferring files between computers

(c) Describe how **packet switching** is used to send a file across a network. **[4]**

(d) Explain why a packet may arrive at its destination out of order. **[2]**

7 The TCP/IP model has four layers.

(a) Name the **four** layers of the TCP/IP model. **[4]**

(b) State the purpose of layering in network communications. **[2]**

(c) A web browser asks for a web page. State the name of the layer where HTTP works. **[1]**

(d) What is meant by a **network standard**? **[1]**

8 Servers, clients and the internet.

(a) State what is meant by a **client-server network**. **[2]**

(b) Give **two** advantages of a client-server network compared with a peer-to-peer network. **[2]**

(c) Give **one** disadvantage of a client-server network. **[1]**

(d) Describe how a peer-to-peer network works. **[2]**

(e) A company stores its files in the **cloud**. Give **two** advantages of this. **[2]**

(f) Give **one** disadvantage of using cloud storage. **[1]**

9 Extended response questions.

(a) *A small business with 15 staff is setting up a new office network. It will include a file server, printers, and staff laptops used throughout the building. Discuss the hardware and network design the business should use. You should consider topology, wired and wireless connections and security. **[8]**

(b) *Explain how data is sent from a computer in the UK to a web server in another country when a user clicks on a link in a web browser. Refer to protocols, packets and addressing in your answer. **[8]**

END OF QUESTIONS

MARK SCHEME – Topic 1.3: Networks, connections and protocols

How to use this mark scheme: mark your own work in a different colour. A bullet (•) in the Answer column is a separate marking point worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed. **Accept** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. For programming questions, any valid solution that works is acceptable and syntax errors are not penalised if the intent is clear. The code in the answers uses the OCR Reference Language style.

Question	Answer	Marks	Guidance
1(a)	• LAN covers a small geographical area (e.g. one building / site) ; • WAN covers a large geographical area (e.g. cities, countries) ;	2 AO1 1.3.1	Accept: LANs are usually owned by the organisation; WANs often use leased lines / other providers.
1(b)	the internet ;	1 AO1 1.3.1	Accept: a bank's network linking branches in different cities.
1(c)	• share files / resources (e.g. printers) ; • central storage / backup / central management of software and users ; • users can log in from any computer / communicate ;	2 AO1 1.3.1	Any two.
1(d)	• bandwidth ; • number of devices / users connected ; • transmission media (wired / wireless) ; • error rate / interference ; • network traffic / quality of hardware ;	2 AO1 1.3.1	Any two.
1(e)	• more data / traffic is sent over the network ; • the available bandwidth is shared so each user receives less / data collisions / delays ;	2 AO2 1.3.1	
2(a)	switch ;	1 AO2 1.3.1	Do not accept hub.
2(b)	• if one cable / device fails the others keep working ; • data collisions are reduced / better performance as each device has its own connection ; • easy to add new devices without disrupting the network ; • easier to find faults ;	2 AO1 1.3.1	Any two.
2(c)	• if the central switch fails the whole network stops ; • more cable required / expense (extra switch) ;	1 AO1 1.3.1	Any one.
2(d)	• connects the local network to other networks / the internet ; • directs / forwards data packets to the correct destination (using IP addresses) ;	2 AO1 1.3.1	
2(e)	network interface card / NIC ;	1 AO1 1.3.1	Accept: network adapter.
2(f)	wireless access point / WAP ;	1 AO2 1.3.1	
3(a)	4 ;	1 AO2 1.3.1	
3(b)	• data can be sent along alternative routes so network keeps working if a link / device fails ; • more reliable / less chance of failure of the whole network ; • faster transfer / less congestion as traffic can be shared across routes ;	2 AO1 1.3.1	Any two.

Question	Answer	Marks	Guidance
3(c)	• expensive / needs more cable and hardware ; • complex to set up and maintain ;	1 AO1 1.3.1	Any one.
3(d)	• A to C or A to B ; • then to D (A → B → D or A → C → D) ;	2 AO3 1.3.1	
4(a)	• devices can move around / mobile ; • no cables needed so cheaper installation / easier to add devices ; • more devices can connect without extra ports ;	2 AO1 1.3.2	Any two.
4(b)	• slower / less bandwidth ; • signal can be affected by interference / walls / distance ; • less secure (data can be intercepted) ;	2 AO1 1.3.2	Any two.
4(c)	Bluetooth ;	1 AO1 1.3.2	
4(d)	Ethernet / copper (twisted pair) / fibre optic ;	1 AO1 1.3.1	Accept: coaxial; CAT5e / CAT6.
4(e)	• wired connection (e.g. fibre optic / Ethernet) ; • faster / higher bandwidth ; • more reliable / less interference / more secure ;	3 AO3 1.3.2	
4(f)	• data is scrambled using an algorithm / key (turned into ciphertext) ; • so it cannot be understood if it is intercepted ; • only users with the correct key can decrypt / read it ;	3 AO2 1.3.2	
5(a)	192.168.1.24	1 AO1 1.3.2	
5(b)	• IP address is assigned by the network / can change, MAC address is fixed by the manufacturer / permanent ; • IP address works at network level / used to route data across networks, MAC address identifies the NIC on the local network ;	1 AO1 1.3.2	Must be a clear comparison.
5(c)	• IPv4 only has about 4.3 billion addresses, not enough for all devices ; • IPv6 gives many more addresses (128-bit) ;	2 AO1 1.3.2	
5(d)	uniquely identifies a device / network card on a network ;	1 AO1 1.3.2	
5(e)	• 1: the browser sends the domain name to the DNS server ; • 2: the DNS server returns the matching IP address ; • 3: the browser sends a request to the web server at that IP address ; • 4: the web server sends the web page back to the browser ;	4 AO2 1.3.2	
6(a)	a set of rules for how data is sent / received / formatted between devices ;	1 AO1 1.3.3	
6(b)	• HTTP: transferring web pages ; • HTTPS: transferring web pages securely (encrypted) ; • FTP: transferring files between computers ; • SMTP: sending email ; • IMAP: retrieving email while keeping it on the server ;	5 AO1 1.3.3	One mark for each correct match.

Question	Answer	Marks	Guidance
6(c)	• the file is split into packets ; • each packet has a header with the destination address / packet number ; • packets can take different routes across the network ; • at the destination the packets are put back in order / reassembled ;	4 AO1 1.3.3	
6(d)	• packets travel by different routes ; • some routes are faster / more congested than others ;	2 AO2 1.3.3	
7(a)	• application ; • transport ; • internet ; • link / network access ;	4 AO1 1.3.3	
7(b)	• breaks the network into smaller parts / each layer has its own job ; • layers can be changed / developed independently ; • helps manufacturers make compatible hardware / easier to find faults ;	2 AO1 1.3.3	Any two.
7(c)	application ;	1 AO2 1.3.3	
7(d)	an agreed set of rules / specifications (e.g. for hardware / protocols) so that different manufacturers' devices work together ;	1 AO1 1.3.3	
8(a)	• one or more central servers provide services / data / resources ; • clients (user computers) request these services from the server ;	2 AO1 1.3.1	
8(b)	• central backup of files ; • central management of security / user accounts / software updates ; • more reliable / easier to manage large numbers of users ;	2 AO1 1.3.1	Any two.
8(c)	• server is expensive to buy / maintain ; • if the server fails clients cannot access resources ;	1 AO1 1.3.1	Any one.
8(d)	• computers are connected directly with equal status ; • each computer can share its own files / resources and no central server is required ;	2 AO1 1.3.1	
8(e)	• files can be accessed from anywhere with an internet connection ; • no need to buy / maintain own servers / storage can be scaled easily ; • provider carries out backups ;	2 AO1 1.3.1	Any two.
8(f)	• needs an internet connection ; • security / privacy concerns / provider can be hacked or fail ; • ongoing subscription cost ;	1 AO1 1.3.1	Any one.

Question	Answer	Marks	Guidance
9(a)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • star topology with a central switch: each device has its own connection, so a fault with one cable does not affect the others; easy to expand; but the switch is a single point of failure • router to connect the office to the internet; NIC in each device • file server wired with Ethernet (fast, reliable, secure) as it carries a lot of traffic; printers wired or wireless • wireless access points for laptops so staff can move around the building; slower and more prone to interference • client-server model for central file storage, backups and user accounts, compared with peer-to-peer • security: encrypt wireless traffic (WPA2/3), firewalls, strong passwords, access rights, anti-malware • balanced judgement about cost versus performance / reliability	8 AO1 / AO2 / AO3 1.3.1 / 1.3.2	

Question	Answer	Marks	Guidance
9(b)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • browser needs the IP address of the web server; domain name sent to a DNS server which returns the IP address • request uses HTTP / HTTPS at the application layer • the data is split into packets by TCP (transport layer) • each packet has a header with the source and destination IP addresses and a packet number • routers read the destination IP address and forward each packet towards the destination; packets may take different routes • MAC addresses are used to move packets between devices on each local network (link layer) • at the web server the packets are reassembled in the correct order using the packet numbers and any missing / corrupt packets are re-requested • the web server replies in the same way, sending the web page back to the user's computer	8 AO1 / AO2 1.3.2 / 1.3.3	

Where marks are lost in Topic 1.3

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Topology questions:** always compare. 'Star is more reliable' needs the reason: 'if one cable fails the others keep working'.
- ☐ **Switch vs router:** a switch connects devices *within* a LAN; a router connects *different networks* (e.g. LAN to internet).
- ☐ **IP vs MAC:** IP can change and routes data between networks; MAC is permanent and identifies the NIC.
- ☐ **Protocol mix-ups:** SMTP = *send* email; IMAP / POP = *receive*; HTTPS = HTTP + encryption; FTP = files.
- ☐ **Packet switching:** split → add headers → different routes → reassemble in order. Four marking points every time.
- ☐ **Encryption:** scramble using a key so intercepted data is unreadable. Do not say it 'stops hackers getting the data'.
- ☐ **Wired vs wireless:** link advantages to the scenario (mobility, cost) not just 'faster'.
- ☐ **Discuss questions:** weigh each design choice, then give a justified recommendation.